B. Intersection

time limit per test: 1 second
memory limit per test: 256 megabytes

You are given two set of points. The first set is determined by the equation $A_1x + B_1y + C_1 = 0$, and the second one is determined by the equation $A_2x + B_2y + C_2 = 0$.

Write the program which finds the number of points in the intersection of two given sets.

Input

The first line of the input contains three integer numbers A_1, B_1, C_1 separated by space. The second line contains three integer numbers A_2, B_2, C_2 separated by space. All the numbers are between -100 and 100, inclusive.

Output

Print the number of points in the intersection or `-1` if there are infinite number of points.

Examples

input Copy

```
1 1 0
2 2 0
```

output Copy

```
-1
```

input Copy

```
1 1 0
2 -2 0
```

output Copy

```
1
```

Codeforces Alpha Round 21 (Codeforces format)

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, msys2

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
371469064	Apr/17/2026 11:09	Accepted
371468591	Apr/17/2026 11:04	Wrong answer on test 9
371468425	Apr/17/2026 11:02	Wrong answer on test 121
371462937	Apr/17/2026 09:50	Wrong answer on test 121

→ Problem tags

implementation math *2000

No tag edit access

→ Contest materials

- Announcement

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Server time: Apr/17/2026 15:17:49 UTC+7 (n2).
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